Air Quality and Income Inequality: Canadian Metropolitan Areas

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The complex dynamics between climatic and social systems shape the environment and the economy. Environmental exposures have a particularly unmistakable impact on urban centers: air pollution is widespread and results in adverse health effects. Previous literature has concluded that stringent air quality regulations can exacerbate income inequality (Jha et al., 2019). However, since areas with high air pollution are often subject to especially strict regulation, there may instead be a relationship between poor air quality and income inequality. This paper evaluates this relationship within Canadian Metropolitan Areas (CMAs). Using Air Quality Health Index (AQHI) data, spatial data, and demographic data, we construct an Ordinary Least Squares regression to determine if there is a statistically significant relationship between AQHI and Gini Coefficient. We find a significant, positive relationship between Average AQHI and Gini Coefficient within Canadian CMAs ($\beta_1 = 0.089$, $p = 0.0013$). Additionally, the effect of Average AQHI on Gini Coefficient decreases in CMAs with a higher mean income ($\beta_{2a} = -0.035$, $p = 9.994 \times 10^{-5}$; $\beta_{2b} = -0.064$, $p = 3.437 \times 10^{-11}$). These findings suggest both that air pollution is associated with higher income inequality in CMAs and that air pollution has a weakened effect on income inequality in wealthier CMAs. By illustrating this relationship, we hope to contribute to a robust understanding of the connection between environmental exposures and human outcomes. Assessing and understanding the interactions between climatic and social systems allows for the implementation of more accurate, equitable, and effective environmental policy.

As climate change reshapes environmental regimes worldwide, academic and political interest in the interactions between climatic and social systems has grown. Environmental exposures, including air pollution, are understood to impact human outcomes (Burke et al., 2022). Studies which evaluate the varied effects of environmental exposures across social and economic subgroups are especially salient (Landrigan et al., 2018). However, the impact of environmental exposures on urban economic conditions is underexplored.

Air pollution is a particularly important source of environmental exposure: globally, "approximately 86% (2.5 billion inhabitants) of urban inhabitants lived in urban areas that exceeded WHO's 2005 guideline annual average $PM_{2.5}$" (Southerland et al., 2022). Previous research has determined that stringent air quality regulations, such as those instated by the Clean Air Act, can exacerbate income inequality (Jha et al., 2019). This conclusion may cause regulatory agencies to avoid implementing environmental policy in order to avoid negative economic outcomes. Under the Clean Air

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Act, a county that is out of attainment with the American National Ambient Air Quality Standards is subject to "stricter environmental regulations than attainment counties" (Jha et al., 2019).

This detail leads to a potential misattribution of causality in the relationship between strict air quality standards and income inequality. If urban areas with poor air quality are less resilient to increases in income inequality, it could be poor air quality—not air quality regulation—which is associated with higher relative income inequality. In principle, this destabilizing effect is possible. Poor air quality has a disproportionately negative impact on low-income households (Deryugina et al., 2021). Additionally, poor air quality is associated with worsened labor productivity due to adverse health impacts (Office of Air and Radiation, 2011). Together, these impacts on the workforce might worsen or prolong the short- and long-term impacts of an economic shock on income inequality.

A first step to determining whether air pollution has a destabilizing effect on income equality is to determine whether a relationship between air quality and income inequality exists. **This paper asks if air quality has a statistically significant effect on income inequality within Canadian Metropolitan Areas (CMAs)**. If the social consequences of poor air quality worsen an economy's resilience to income inequality, we would find higher income inequality in CMAs with generally higher air pollution. As such, **we expect to find a positive relationship between air pollution and income inequality**.

Our results suggest that this hypothesis was correct: we find a significant, positive relationship between Average AQHI and Gini Coefficient within Canadian CMAs ($\beta_1 = 0.089$, $p = 0.0013$). Additionally, the effect of Average AQHI on Gini Coefficient decreases in CMAs with a higher mean income ($\beta_{2a} = -0.035$, $p = 9.994 \times 10^{-5}$; $\beta_{2b} = -0.064$, $p = 3.437 \times 10^{-11}$). This suggests that air pollution is associated with higher income inequality in Canadian Census Metropolitan Areas, and that air pollution has a weakened effect on income inequality in wealthier Census Metropolitan Areas. By illustrating this relationship, we hope to contribute to a robust understanding of the connection between environmental exposures and human outcomes. Assessing the complex interactions between climatic and social systems allows for the implementation of more accurate, equitable, and effective environmental policy.

I. Data Description

All of the data used in this research was collected from one of three sources: Environment and Climate Change Canada, Statistics Canada's 2021 Census, and Environics Analytics' 2024 Demographic Estimates & Projections.

Air Quality Health Index (AQHI) is a metric of air pollution created by Environment and Climate Change Canada. We have chosen CMA Average AQHI as our independent variable of interest. Using a risk-weighted scale of ground-level ozone (O_3), particulate matter ($PM_{2.5}$ & PM_{10}) and nitrogen dioxide (NO_2), the index provides a value typically ranging between 1 and 10 (Government of Canada, 2021). **Higher AQHI values indicate higher air pollution.** AQHI values between 1 and 3 are associated with low health risk; values between 4 and 6 are associated with a moderate health risk; values between 7 and 10 are associated with a high health risk; and values above 10 indicate a very high health risk (Government of Canada, 2015). The AQHI index is typically simplified and reported to the public as integer values—but hourly observation data accurate to two decimal places from December 2023 until April 2025 is currently available. Hourly data is organized into CSV files by month of observation and geographic region. Due to provincial policy, no AQHI data from Quebec is publicly available (Environment and Climate Change Canada, 2025). We collected all AQHI observations from 2024 and combined each measurement station's hourly data into a daily average AQHI. We also collected a GeoJSON file which contains spatial data about every AQHI measurement station (Environment and Climate Change Canada, 2025). This GeoJSON file was converted to the SHP file format.

To associate this AQHI data with corresponding CMAs, we collected provincial and CMA spatial data files from Statistics Canada's 2021 Census data (Statistics Canada, 2021a; Statistics Canada, 2021b). This data is a set of boundary measurement SHP files which were last updated in 2021. This is acceptable: CMA boundaries have not changed since the 2021 census. In addition to providing geographic data, these boundary sets include a land area measurement (in km^2) and Census Population Designation for each CMA. All CMAs have one of three Census Population Designations: "B," a CMA with a population >100,000 with >50,000 living in a designated CMA core ($n = 34$); "D," an untracted Census Agglomeration with a designated core population >10,000 ($n = 23$); and "K," a tracted Census Agglomeration with a designated core population >50,000 ($n = 6$) (Statistics Canada, 2018). We adjusted the Coordinate Reference Systems of all relevant geographic data to NAD 83 projected to WGS 84 (SRID: 3347). We were then able to group CMAs with their AQHI measurement stations. Not all AQHI measurement stations are located within a CMA, and not all CMAs contain an AQHI measurement station—these incomplete datasets are excluded from our study. Additionally, when a CMA contained multiple AQHI measurement stations, their daily averages were combined. Having organized our daily average AQHI data by CMA, we then calculated each CMA's Average AQHI in 2024. Additionally, we calculated each CMA's percentage of daily average AQHI values greater than 3 (indicating moderate health risk or higher). Days without any AQHI reading were omitted from the calculation of these averages. One CMA, Brandon, was ommitted because it had less than 100 days of collected AQHI data: this could introduce a potential bias in AQHI measurement, as air pollution varies across seasons (Zareba and Weglinska, 2024).

Demographic data about every CMA was collected from Environics Analytics' 2024 Demographic Estimates & Projections (Environics Analytics, 2024). Official census data for 2024 does not exist, but this dataset—which was created "in conjunction with experts at The Centre for Spatial Economics"—is one of the best available projections of Canadian demographic data (Environics Analytics, 2024). It was derived from Statistics Canada's 2021 Census data and made available through the SimplyAnalytics platform. We chose to collect the following 2024 data about every CMA:

1) Mean Income ($)
2) Median Household Income ($)
3) Percent of Households with a Household Income $100,000 or Over (%)
4) Percent of Households with a Household Income of $100,000 or Over with a Household Income of $200,000 or Over (%)
5) Percent of Households with a Household Income of $200,000 or Over with a Household Income of $300,000 or Over (%)
6) Percent Male Population (%)
7) Percent of Population with Visible Minority Status (%)
8) Percent of Population >15 Years Old Married or Living With A Common-Law Partner (%)
9) Percent of Population without Knowledge of English or French (%)
10) Percent Immigrant Population (%)
11) Employment Rate (%)
12) Percent of Population >15 Years Old in the Labour Force (%)

The inclusion and exclusion of this data in our regression model will be explained and evaluated further in Section III.

The CMA demographic data does not include a measurement of income inequality. The Gini Coefficient is a measure of income inequality $Gini \in [0, 1]$, with $Gini = 0$ representing perfect

income equality and $Gini = 1$ representing maximum income inequality. We have chosen the Gini Coefficient as our dependent variable. We estimated the Gini Coefficient for each CMA population using the following vector $\vec{I_i}$ of 2024 demographic data:

$$\vec{I_i} = \begin{bmatrix} I_0\text{: Median Household Income} \\ I_1\text{: Percent of Households with a Household Income \$100,000 or Over} \\ I_2\text{: Percent of Households with a Household Income \$100,000 or Over with a Household Income of \$200,000 or Over} \\ I_3\text{: Percent of Households with a Household Income \$200,000 or Over with a Household Income of \$300,000 or Over} \end{bmatrix}$$

From $\vec{I_i}$ we can approximate the Lorenz Curve and corresponding Gini Coefficient of every CMA. We used a monotonically increasing cubic spline interpolation to generate a polynomial representing the Lorenz Curve of every CMA. The Gini Coefficient was then approximated through numerical integration. The complete approximation process is detailed in Appendix A.

Two CMAs were excluded from the dataset because of their estimated Gini Coefficients: Wood Buffalo and Yellowknife. Both CMAs have very high mean incomes, which means that our upper bound of household income ($300,000) is not sufficient to accurately estimate the concavity of the Lorenz Curve.

No noticable outliers are present in the remainder of our CMA demographic data. All other listed variables acted as control variables, and their purpose will be discussed further in our Model Specification section.

II. Summary Statistics

Table 1 provides descriptive statistics for the primary variables employed in our empirical analysis. The table summarizes data from CMAs across Canada, including measures of income inequality, air quality, demographic characteristics, and CMA land area. The mean Gini Coefficient is 0.383 with a standard deviation of 0.100, reflecting variation in income inequality across CMAs. The mean Average AQHI is 2.054 ($s = 0.287$), and the mean percentage of high AQHI days is 19.44% (s = 21.932).

It is helpful to understand the simple correlation between each of these variables: high correlation values could be an indicator of multicollinearity. Table 2 displays the correlation between all considered variables.

We began by estimating a bivariate linear regression of Average AQHI on Gini Coefficent as displayed in Figure 1.

III. Model Specification

A. Main Model Specification

The core objective of this model is to estimate the effect of air quality (as captured by AQHI) on income inequality (as measured by the Gini Coefficient) within CMAs. To investigate the relationship between air quality and income inequality, we employ an ordinary least squares (OLS) regression model with the functional form:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i \times IncomeCat_i + \sum_{k=1}^{6} (\beta_{k+2} W_{k,i}) + \epsilon_i \tag{1}$$

The dependent variable in this model is $Gini_i$, our estimation of the Gini Coefficient. The independent variable of interest is $AQHI_i$, which is the average AQHI in a given CMA. This captures

Table 1—: Census Metropolitan Areas: Key Variables ($n = 63$)

Variable	**Mean**	**S.D.**	**Min.**	**Max.**
Gini Coefficient	0.383	0.100	0.142	0.586
Average AQHI	2.054	0.287	1.455	2.689
High AQHI Days (%)	19.444	21.932	0.000	85.000
Average Income ($)	56,564.91	7,484.71	43,216.52	77,301.47
Male Population (%)	49.713	0.854	48.156	52.114
Visible Minority (%)	15.394	12.093	2.208	58.503
Married or Common-law (%)	56.331	2.444	48.760	61.884
No English or French (%)	0.902	1.091	0.074	5.639
Immigrant (%)	15.888	9.435	3.017	49.748
Labour Force (%)	51.357	4.267	42.844	62.069
Employment Rate (%)	91.359	2.212	82.300	96.600
Land Area (km^2)	3491.412	4223.093	54.843	21708.620

Note: S.D. represents standard deviation. Min. represents the minimum value of a variable within our CMA data. Max. represents the maximum value of a variable within our CMA data.
Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

Table 2—: Census Metropolitan Areas: Variable Correlation ($n = 63$)

Variable	**G.C.**	**A.A.**	**H.A.D.**	**A.I.**	**M.P.**	**V.M.**	**M.C.**	**N.E.F.**	**I.**	**L.F.**	**E.R.**	**L.A.**
Gini Coefficient	—	-0.224	-0.267	-0.879	-0.551	-0.475	-0.277	-0.302	-0.527	-0.397	-0.726	-0.018
Average AQHI		—	0.907	0.377	0.237	0.359	0.019	0.212	0.361	-0.023	0.230	0.004
High AQHI Days (%)			—	0.407	0.288	0.393	0.085	0.233	0.431	-0.009	0.244	0.137
Average Income ($)				—	0.471	0.360	0.329	0.114	0.405	0.363	0.650	0.098
Male Population (%)					—	0.192	0.240	0.178	0.185	0.184	0.461	0.189
Visible Minority (%)						—	0.042	0.880	0.971	0.135	0.550	-0.027
Married or Common-law (%)							—	0.006	0.083	0.324	0.362	0.034
No English or French (%)								—	0.835	0.089	0.354	-0.051
Immigrant (%)									—	0.232	0.569	-0.047
Employment Rate (%)											0.420	-0.153
Labour Force (%)											—	-0.032
Land Area (km^2)												—

Note: All values listed are the simple correlation $\rho_{X,Y}$ of the associated row and column variables. The correlation matrix is symmetric. G.C. represents Gini Coefficient. A.A. represents Average AQHI. H.A.D. represents High AQHI Days (%). A.I. represents Average Income ($). M.P. represents Male Population (%). V.M. represents Visible Minority (%). M.C. represents Married or Common-law (%). N.E.F represents No English or French (%). I. represents Immigrant (%). E.R. represents Employment Rate (%). L.F. represents Labour Force (%). L.A. represents Land Area (km^2).
Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

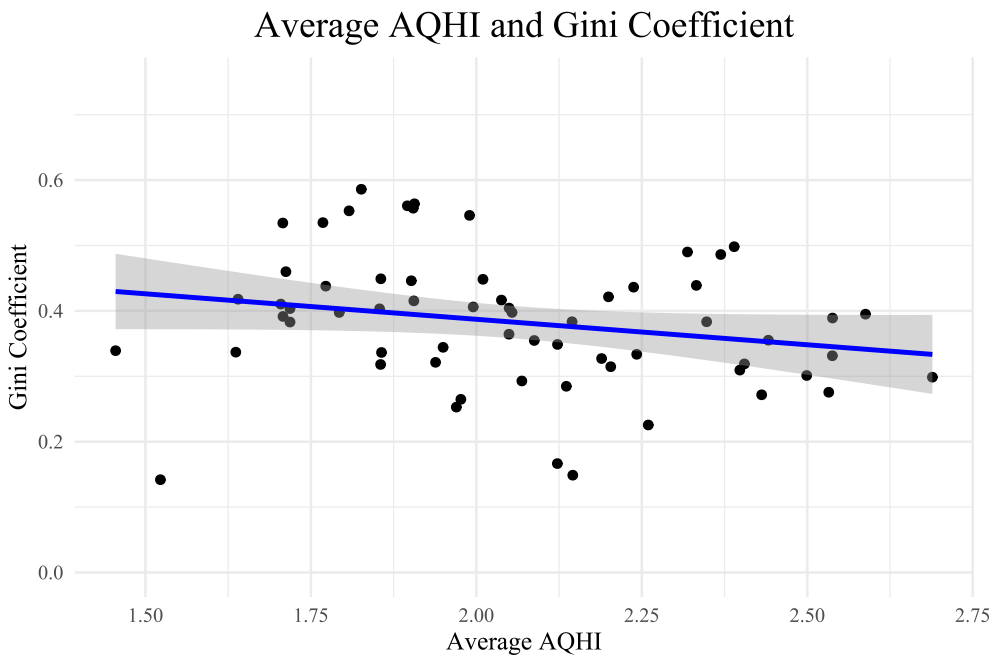


Figure 1. : Scatterplot and Bivariate Regression of CMA Average AQHI and Gini Coefficient

Note: Scatterplot represents CMA data values for Average AQHI and Gini Coefficient. The blue line represents the simple bivariate regression of the form $\hat{Y} = \beta_0 + \beta_1 X_i$ where $\hat{Y}$ represents predicted Gini Coefficient, β_0 represents the predicted intercept, β_1 represents the coefficient on Average AQHI, and X_i represents a compact space of Average AQHI values such that $X_i \in [1.455, 2.689]$: the range of our collected values of Average AQHI. β_0 had an estimated value of 0.543 with a standard error of 0.090. β_1 had an estimated value of -0.078 with a standard error of 0.043.
Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

typical air quality conditions, accounting for various types of pollutants (Government of Canada, 2021). This model will directly test whether *ceteris paribus* differences in air quality are associated with changes in the Gini Coefficient. For example, if the estimated coefficient β_1 on $AQHI_i$ is significant and positive, that would indicate that worse air quality is correlated with higher income inequality within a CMA. We also have included an interaction variable constructed from $AQHI_i \times IncomeCat_i$, which is a categorical variable representing the mean income for a given CMA. We include this variable because we expect air quality to effect income inequality differently in places with higher or lower average incomes due to the heterogenous health impacts of air pollution (Deryugina et. al, 2021). Adding in this interaction variable will not introduce endogeneity: mean income was not used in our calculation of the Gini Coefficient, and the interaction variable only accounts for changes in the effect of $AQHI_i$ on $Gini$ across different levels of average CMA income. Since we are not including average income directly, we have centered and categorized CMA mean incomes. This will make our interpretation of the model more clear and avoid any problems caused by potential non-linearity in the interaction variable. The variable $IncomeCat$ takes on three values: $IncomeCat_{Low}$, corresponding to CMAs with a mean income in the bottom 33% of all studied CMAs; $IncomeCat_{Mid}$, corresponding to CMAs with a mean income in the middle 33% of all studied CMAs; and $IncomeCat_{High}$, corresponding to CMAs with a mean income in the top 33% of all studied CMAs.

We have chosen the following control variables:

1) Percent Male Population (%)
2) Percent of Population >15 Years Old Married or Living With A Common-Law Partner (%)
3) Percent Immigrant Population (%)
4) Percent of Population >15 Years Old in the Labour Force (%)
5) Employment Rate (%)
6) CMA Land Area (km^2)

Percent male population is included as a control variable because there is higher male representation in industries like manufacturing, which are correlated with both air pollution and income inequality (Alexikova et al., 2016).

Marital status is included as a control variable: in lower-income regions, dual-income households often exhibit higher financial stability than single-income households (Downey et al., 2017).

Percent immigrant population is included as a control variable because recent immigrants tend to concentrate in low-income, high-pollution areas (American Lung Association, 2023). Additionally, increases in income inequality in Canada have historically been associated with immigrant populations (Picot and Hou, 2019).

We include labor force participation rate and employment rate to account for variations in economic prosperity across CMAs.

CMA land area is included as a control variable to address potential heterogeneity between CMAs caused by geographic fixed effects. Geographically larger CMAs might vary in both pollution levels and income inequality; standardizing air quality and income inequality by land area adjusts for this fixed effect.

We ultimately decided to omit two control variables to account for high multicollinearity. The percentage of immigrants in total population has an extremely high correlation with both the percent of population with visible minority status ($\rho = 0.971$) and the percent of population without knowledge of English or French ($\rho = 0.835$). As such, both of these variables were not included in our regression.

We recognize the potential of omitted variables in our model; however, through the inclusion of these control variables, we assume Conditional Mean Independence such that the error term ϵ is not correlated with our independent variable of interest $AQHI$: $E[\epsilon|AQHI] = 0$.

The error term represents unobserved factors influencing local income inequality that are not explicitly included in the model.

B. Model Variations

To ensure the validity of our results, we propose four alternative models.

We first add $AQHI_i^2$ to our regression to account for a possible nonlinear effect of air quality on income inequality. Different levels of environmental exposure to poor air quality might be associated with varying health and economic consequences: by including both a linear and a quadratic term, we can test whether the effects of higher pollutant concentrations follow a simple linear pattern or exhibit diminishing/increasing marginal effects (Banzhaf & Walsh, 2008; Landrigan et. al, 2018). This gives us the following alternative functional form:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i^2 + \beta_3 AQHI_i \times IncomeCat_i + \sum_{k=1}^{6} (\beta_{k+3} W_{k,i}) + \epsilon_i \tag{2}$$

Next, we replace the interaction variable $AQHI_i \times IncomeCat_i$ with the quadratic term $AQHI_i^2$. This change tests whether AQHI's effect on income inequality actually varies in CMAs with higher or lower average incomes: it's possible that a direct nonlinear effect of air quality on income inequality is obscured by the interaction variable. This gives us an alternative functional form:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i^2 + \sum_{k=1}^{6} (\beta_{k+2} W_{k,i}) + \epsilon_i \tag{3}$$

Next, we remove the control variable for CMA land area to verify that our results are not unduly influenced by the inclusion of geographic fixed effects and that we are not overfitting our regression. This gives us the following alternative functional form:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i \times IncomeCat_i + \sum_{k=1}^{5} (\beta_{k+2} W_{k,i}) + \epsilon_i \tag{4}$$

Finally, we group our data by CMA Census Population Designation to capture potential heterogeneity caused by population size: more populous CMAs may vary in both pollution levels and income inequality. This gives us a regression model of the functional form:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i \times IncomeCat_i + \sum_{k=1}^{7} (\beta_{k+2} W_{k,i}) + \epsilon_i \tag{5}$$

IV. Table of Results

Table 3 displays the results of our main functional form, given by Equation (1) and four alternative functional forms, given by Equations (2-5). Significance values of each coefficient estimate are noted.

V. Discussion

A. Results

Our findings suggest that Average AQHI is a significant determinant of Gini Coefficient in Canadian CMAs. In our main functional form, the coefficient estimate β_1 on Average AQHI

Table 3—: Main Functional Form and Alternative Specifications: Determinants of Income Inequality in Canadian CMAs ($n = 63$).

	Dependent Variable: Gini Coefficient				
	Main	Alt. 1	Alt. 2	Alt. 3	Alt. 4
Variable	(1)	(2)	(3)	(4)	(5)
Intercept	2.807***	3.424***	3.591***	2.766***	2.791***
	(0.319)	(0.459)	(0.701)	(0.343)	(0.325)
Average AQHI	0.089**	−0.415	−0.378	0.085**	0.087**
	(0.026)	(0.269)	(0.357)	(0.028)	(0.026)
Average AQHI: Medium Income	−0.035***	−0.033***	—	−0.032***	−0.036***
	(0.008)	(0.008)		(0.008)	(0.009)
Average AQHI: High Income	−0.064***	−0.065***	—	−0.060***	−0.064***
	(0.008)	(0.008)		(0.008)	(0.008)
(Average AQHI)^2	—	0.123	0.093	—	—
		(0.064)	(0.082)		
Male (%)	−0.046***	−0.047***	−0.036**	−0.042***	−0.046***
	(0.007)	(0.007)	(0.012)	(0.008)	(0.008)
Married (%)	0.003	0.002	−0.001	0.003	0.003
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Immigrant (%)	−0.002*	−0.002**	−0.002*	−0.002*	−0.002*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Employment Rate (%)	−0.002	−0.02	−0.005	−0.003	−0.002
	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)
Labour Force (%)	−0.004	−0.004	−0.010***	−0.004	−0.004
	(0.002)	(0.002)	(0.003)	(0.002)	(0.003)
Land Area (km^2)	[+]	[+]	[−]	—	[+]
Grouped by Census Designation	—	—	—	—	✓
R^2	0.802	0.812	0.632	0.793	0.802
Adjusted R^2	0.776	0.764	0.577	0.762	0.760

Note: Robust standard errors are listed in parentheses. Coefficient estimate significance is indicated as follows: *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$. This table presents the results of OLS regressions estimating the relationship between Average AQHI and Gini Coefficient across Canadian Census Metropolitan Areas (CMAs). Average AQHI is the independent variable of interest in all models. The main functional form, shown in Equation (1), includes an interaction variable between Average AQHI and CMA Mean Income. Alternative specifications (2–5) test model robustness. Alt. 1 is the functional form described in Equation (2) which adds (Average AQHI)^2. Alt. 2 is the functional form described in Equation (3) which replaces the interaction variable between Average AQHI and CMA Mean Income with (Average AQHI)^2. Alt. 3 is the functional form described in Equation (4) which omits the Land Area control variable. Alt. 4 is the functional form described in Equation (5) which groups CMAs by Census Designation. The coefficient estimate β_8 for Land Area is non-significant with a magnitude $|\beta_8| < 10^{-5}$ in all functional forms; the sign of the coefficient estimate is noted.

Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

is 0.089 ($p = 0.0013$). This coefficient estimate is statistically significant and positive; holding all other factors constant, a 1-unit increase in Average AQHI in a low-income CMA increases that CMA's expected Gini Coefficient by 0.089.

In all functional forms containing the interaction variable $AQHI_i \times IncomeCat_i$, the coefficient estimates β_{2a} and β_{2b} are statistically significant. In our main functional form, β_{2a} has a coefficient estimate of -0.035 ($p = 9.994 \times 10^{-5}$). Holding all other factors constant, in middle-income CMAs, a 1-unit increase in Average AQHI increases a CMA's expected Gini Coefficient by 0.054. In our main functional form, β_{2b} has a coefficient estimate of -0.064 ($p = 3.437 \times 10^{-11}$). Holding all other factors constant, in high-income CMAs, a 1-unit increase in Average AQHI increases a CMA's expected Gini Coefficient by 0.026.

Our main functional form has a high explanatory power (Adjusted $R^2 = 0.776$). This confirms our hypothesis: there is a significant, positive relationship between air pollution and income inequality. Additionally, the effect of Average AQHI on Gini Coefficient decreases in CMAs with a higher mean income: air pollution is associated with higher income inequality, but air pollution has less of an impact on income inequality in wealthier CMAs. These insights could help inform policy targeting environmental and social inequality. In particular, improving air quality in low-income CMAs could have a meaningful impact on reducing inequality.

All other variables included in the regression are control variables. We will not interpret the magnitude of their effect on estimated Gini Coefficient, but we should evaluate the sign of their coefficient estimates for model robustness:

1) W_1, Male (%): $\beta_3 = [+]$, Expected (Alexikova et al., 2016)
2) W_2, Married (%): $\beta_4 = [+]$, Unexpected (Downey et al., 2017)
3) W_3, Immigrant (%): $\beta_5 = [-]$, Unexpected (American Lung Association, 2023; Picot and Hou, 2019)
4) W_4, Employment Rate (%): $\beta_6 = [-]$, Expected
5) W_5, Labour Force (%): $\beta_7 = [-]$, Expected
6) W_6, Land Area (km^2): $\beta_8 = [+]$, Expected

A positive coefficient estimate on W_1 is expected: there is higher male representation in industries like manufacturing, which are associated with income inequality (Alexikova et. al, 2016).

A positive coefficient estimate on W_2 is unexpected: since dual-income households often exhibit higher financial stability than single-income households, one may expect to see decreased income inequality in CMAs with a higher proportion of dual-income households (Downey et al., 2014). However, the percentage of married or common-law partners in a CMA does not necessarily indicate the number of dual-income households in a CMA. It is possible that a different unknown effect is present.

A negative coefficient estimate on W_3 is unexpected: increases in income inequality in Canada have historically been associated with immigrant populations (Picot and Hou, 2019). However, this may no longer be the case. The proportion of immigrant populations in Canada have continually increased since the 1950s, so it is possible that the relationship between immigrant population and income inequality has changed (Statistics Canada, 2016). CMAs with a high proportion of immigrants may have significant immigrant communities who have developed effective systems of community resilience.

A negative coefficient estimate on W_4 is expected: CMAs with a higher employment rate are likely to have fewer households with no income, thereby decreasing income inequality. Similarly, a negative coefficient estimate on W_5 is also expected: CMAs with a high labour force integration can be expected to have fewer households with no income, thereby decreasing income inequality. Finally, a positive coefficient estimate on W_6 is expected: CMAs with higher land area may have

increased variance in income due to urban sprawl, differences in available opportunities between communities, and higher industrialization.

B. Validation Tests

To assess whether our main functional form is correctly specified with low multicollinearity and heteroskedasticity, we run a Ramsey RESET test, calculate Variance Inflation Factors (VIF), and run a White's test.

For the Ramsey RESET test, our null hypothesis H_0 is that an equation of the functional form

$$(6)\quad Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i \times IncomeCat_i + \sum_{k=1}^{6} (\beta_{k+2} W_{k,i}) + \gamma_1 (\widehat{Gini_i})^2 + \gamma_2 (\widehat{Gini_i})^3 + \epsilon_i,$$

where $\widehat{Gini_i}$ is the predicted value of $Gini_i$ obtained from Equation (1), has coefficient estimates $\gamma_1 = \gamma_2 = 0$. Our alternative hypothesis H_1 is that at least one of γ_1, $\gamma_2 \neq 0$. Performing an F-test, we get $F = 2.0131$ ($p = 0.1441$) and fail to reject the null hypothesis: our functional form is not misspecified.

Since this functional form includes an interaction variable, the R function `vif()` returns generalized VIF values (GVIF). All calculated GVIF values are below 3. All GVIF values are presented in Table 4.

Table 4—: Main Functional Form: Generalized Variance Inflation Factors ($n = 63$)

Variable	**A.A.**	**I.C.**	**M.P.**	**M.C.**	**I.**	**E.R.**	**L.F.**	**L.A.**
GVIF	2.018	2.018	1.469	1.289	1.694	1.382	2.619	1.195
D.F.	3	3	1	1	1	1	1	1

Note: GVIF represents Generalized Variance Inflation Factors. D.F. represents degrees of freedom. A.A. represents Average AQHI. I.C. represents Income Category. M.P. represents Male Population (%). M.C. represents Married or Common-law (%). I. represents Immigrant (%). E.R. represents Employment Rate (%). L.F. represents Labour Force (%). L.A. represents Land Area (km^2).
Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

Constructing our White's test, we have a null hypothesis H_0 that our error term is homoskedastic such that $var\,(u_i|X_i = x) = c$, where $c \in \mathbb{R}$ is a constant. Our alternate hypothesis H_1 is that our error term is heteroskedastic. The result of our White's test, $\chi^2 = 0.12$ ($p = 0.941$), does not indicate heteroskedasticity in our main functional form: we fail to reject the null hypothesis.

VI. Specification Checks

In our first alternate functional form described in Equation (2), we introduced the quadratic term $AQHI_i^2$ to determine whether different levels of Average AQHI are associated with different effects of Average AQHI on Gini Coefficient. From the results of our regression detailed in Table 3, it seems unlikely that this is the case. Adding this quadratic term dramatically impacted the of the coefficient estimate β_1 of $AQHI_I$: β_1 is now negative and nonsignificant (-0.415, $p = 0.1283$). The coefficient estimate β_2 of $AQHI_i^2$ is 0.123 ($p = 0.0602$). This suggests an initially negative, concave upward relationship between $AQHI$ and $Gini$. However, because these results are not significant, we cannot make inferences from this functional form. Additionally, the coefficient estimates β_{3a} and β_{3b} of the interaction variable $AQHI_i \times IncomeCat_i$ were not dramatically impacted by the inclusion of $AQHI_i^2$ (-0.033, $p = 0.0002$; -0.065, $p = 1.421 \times 10^{-11}$). This functional form has less explanatory power than our main functional form (Adjusted $R^2 = 0.764$).

In our second alternate functional form described in Equation (3), we replaced the interaction variable $AQHI_i \times IncomeCat_i$ with the quadratic term $AQHI_i^2$ to determine whether a direct nonlinear effect of air quality on income inequality is obscured by the inclusion of the interaction variable. Once again, adding this quadratic term dramatically impacted the of the coefficient estimate β_1 of $AQHI_I$: β_1 is now negative and nonsignificant (-0.379, $p = 0.2742$). The coefficient estimate β_2 of $AQHI_i^2$ did not dramatically change (0.093, $p = 0.2579$). Because these results are not significant, we cannot make inferences from this functional form. The explanatory power of this functional form is considerably lower than our main functional form and other specifications (Adjusted $R^2 = 0.577$). The interaction variable $AQHI_i \times IncomeCat_i$ seems to substantially contribute to the explanatory power and significance of our model.

In our third alternate functional form described in Equation (4), we removed our control variable for CMA land area to verify that our results are not biased by the inclusion of geographic fixed effects. Removing this variable did not dramatically impact the coefficient estimate β_1 of $AQHI_I$ (0.085, $p = 0.0043$) or the coefficient estimates β_{2a} and β_{2b} of the interaction variable $AQHI_i \times IncomeCat_i$ (-0.031, $p = 0.0002$; -0.060, $p = 2.120 \times 10^{-8}$). This functional form has less explanatory power than our main functional form (Adjusted $R^2 = 0.762$). Removing this control variable only negatively impacts our regression model: our main functional form is more effective.

In our fourth alternate functional form described in Equation (5), we grouped our data by CMA Census Population Designation to capture potential heterogeneity caused by population size. Adding this control variable did not dramatically impact the coefficient estimate β_1 of $AQHI_I$ (0.087, $p = 0.0017$) or the coefficient estimates β_{2a} and β_{2b} of the interaction variable $AQHI_i \times IncomeCat_i$ (-0.036, $p = 0.0001$; -0.064, $p = 2.036 \times 10^{-10}$). This functional form has less explanatory power than our main functional form (Adjusted $R^2 = 0.760$). Adding this control variable only negatively impacts our regression model: our main functional form is more effective.

VII. Robustness Analysis

To further scrutinize the accuracy of our main regression model, we evaluate it through a quantile regression with $\tau = 0.5$.

Quantile regression is less sensitive to outliers, thereby allowing us to see how coefficient estimates differ at various percentiles of the conditional Gini Coefficient distribution. By focusing on the median of $Gini$ by using $\tau = 0.5$, we can evaluate whether the effect of $AQHI_i$ on $Gini$ remains accurate near the middle of the distribution. If the estimated coefficients on $AQHI_i$ and $AQHI_i \times IncomeCat_i$ remain significant and keep the same sign in the quantile regression as calculated by Equation (1), this would affirm that the estimated relationship between $AQHI_i$ and $Gini$ is not merely driven by outliers or skewed distributions in either extreme (e.g., extremely unequal or extremely equal CMAs).

While the coefficient estimate of β_1 on $AQHI_i$ shifts modestly (-0.008), its sign remains positive. Additionally, the 95% confidence interval constructed around the coefficient estimate of β_1 does not include $\beta_1 = 0$, suggesting that this coefficient estimate remains significant. Additionally, the coefficient estimates of β_{2a} and β_{2b} on $AQHI_i \times IncomeCat_i$ have retained their sign and are significant.

This robustness check suggests that the relationship between $AQHI_i$ and $Gini$ is not merely driven by outliers or skewed distributions. Our main model, described in Equation (1), remains an effective estimation of the relationship between $AQHI_i$ and $Gini$.

VIII. Conclusion

This study finds a significant, positive relationship between Average AQHI and Gini Coefficient within Canadian CMAs ($\beta_1 = 0.089$, $p = 0.0013$). Additionally, the effect of Average AQHI on Gini Coefficient decreases in CMAs with a higher mean income ($\beta_{2a} = -0.035$, $p = 9.994 \times 10^{-5}$;

Table 5—: Quantile Regression:
Determinants of Income Inequality in Canadian CMAs ($n = 63$).

	Dependent Variable: Gini Coefficient		
Variable	Coefficient Estimate	Lower Bound	Upper Bound
Intercept	2.906	1.778	3.064
Average AQHI	0.081	0.14	0.131
Average AQHI: Medium Income	−0.025	−0.054	−0.014
Average AQHI: High Income	−0.056	−0.069	−0.038
Male (%)	−0.046	−0.052	−0.030
Married (%)	0.004	0.001	0.007
Immigrant (%)	−0.002	−0.003	−0.001
Employment Rate (%)	−0.004	−0.008	0.001
Labour Force (%)	−0.002	−0.008	0.001
Land Area (km^2)	[+]	[+]	[+]

Note: This table shows the results of a quantile regression using the functional form described in Equation (1) and $\tau = 0.5$. Lower Bound and Upper Bound refer to a 95% confidence interval constructed around the coefficient estimate.

Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

$\beta_{2b} = -0.064$, $p = 3.437 \times 10^{-11}$). **This suggests that air pollution is associated with higher income inequality in Canadian Census Metropolitan Areas, and that air pollution has a weakened effect on income inequality in wealthier Census Metropolitan Areas.**

These findings suggest that there is a possible destabilizing effect of air pollution on income equality. This brings into question the conclusions argued by Jha et al. (2019). Stringent air quality regulations may only be associated with heightened income inequality because air pollution worsens or prolongs the effect of an economic shock on income inequality. The findings of this paper demand further research: future works should seek to identify the relationship between air pollution and an economy's resilience to income inequality. Future studies could employ panel data to evaluate this relationship, establish causality, and identify the mechanisms through which poor air quality contributes to heightened inequality.

Additionally, these findings indicate that air quality regulation may improve—rather than exacerbate—income inequality. Additional environmental intervention in Canadian Census Metropolitan Areas with high Average AQHI readings could help mitigate income inequality.

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A: Mathematical Analysis

We begin with the following vector $\vec{I_i}$ of 2024 demographic data:

$$\vec{I_i} = \begin{bmatrix} I_0\text{: Median Household Income} \\ I_1\text{: Percent of Households with a Household Income \$100,000 or Over} \\ I_2\text{: Percent of Households with a Household Income \$100,000 or Over with a Household Income of \$200,000 or Over} \\ I_3\text{: Percent of Households with a Household Income \$200,000 or Over with a Household Income of \$300,000 or Over} \end{bmatrix}$$

Let us first standardize our data. Let N represent the number of CMAs. For each CMA $k|k \in [1, N]$, we will expand our income vector I_k to a matrix IP_k which contains median-adjusted income and income percentile.

$$IP_k = \begin{bmatrix} 0 & 0.5 & \frac{I_1}{100} & \frac{I_1 I_2}{10000} & \frac{I_1 I_2 I_3}{1000000} \\ 0 & I_0 & \frac{100000}{I_0} & \frac{200000}{I_0} & \frac{300000}{I_0} \end{bmatrix}$$

The first row of IP_k represents the percentile of a given median-adjusted income value with terms x_n. The second row of IP_k represents the median-adjusted income value with terms y_n.

To approximate the Lorenz Curve, we construct a monotonically increasing polynomial $P(x)$ interpolating the set of points (x_n, y_n) with $n \in [0, 4]$. We use the R function `splinefun()` with `method = "hyman"` to create this polynomial using cubic spline interpolation. We also have a median-adjusted line of income equality of the form $E(x) = \frac{y_5}{x_5}x$.

Our Gini coefficient can now be estimated by:

$$G(IP_k) = \frac{\int_0^{x_5} \left(E(x) - P(x)\right)\, dx}{\int_0^{x_5} E(x)\, dx}$$

We find this value using the R formula `integrate()`.